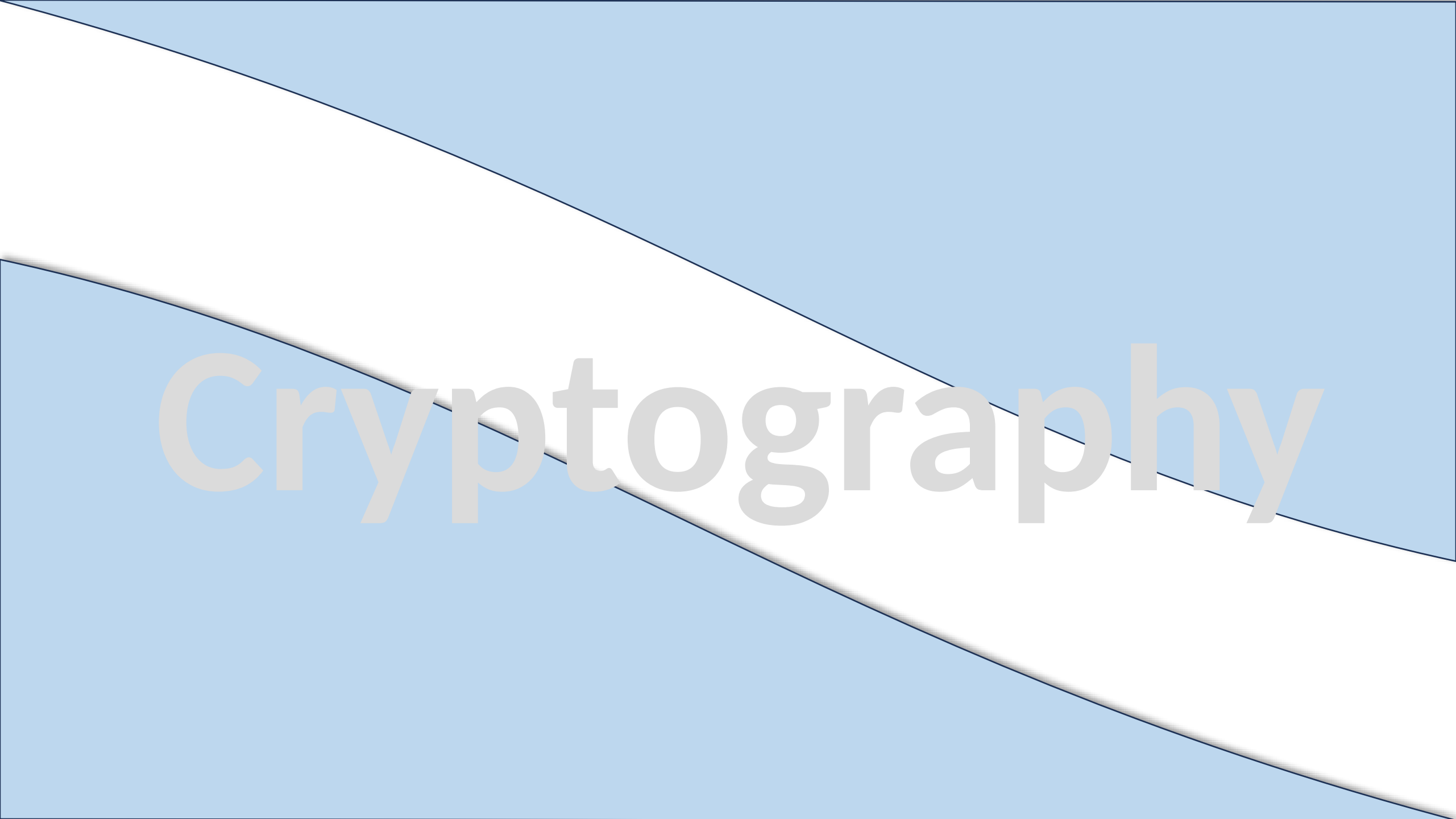


The background consists of two diagonal stripes. The top stripe is white and the bottom stripe is a light blue color. Both stripes have a thin dark blue outline. The word "Cryptography" is centered horizontally and partially overlaps the white stripe.

Cryptography

Cryptography

The study of encoding and decoding secret messages is called cryptography.

Hill Cipher

The **Hill Cipher** is a classical encryption technique based on linear algebra that uses matrix multiplication to encrypt blocks of text. It transforms plaintext into ciphertext using a key matrix and modular arithmetic, typically modulo 26 for the English alphabet.

The **Hill Cipher** was introduced by **Lester S. Hill**, an American mathematician, in **1929**. He developed it as a method to apply linear algebra and modular arithmetic to cryptography, making it one of the first ciphers to use matrix operations for encryption.

Hill Cipher Encryption Steps (2×2 Matrix)

1. Convert the plaintext into uppercase letters
2. Remove spaces and pad the text (if needed) to ensure it has an even number of letters (since we're using 2-letter blocks).
3. Convert letters to numbers using the rule: $A = 1, B = 2, C = 3, \dots, Z = 26$
4. Group the numbers into column vectors of size 2. For HELP $P_1 = \begin{bmatrix} 8 \\ 5 \end{bmatrix}, P_2 = \begin{bmatrix} 12 \\ 16 \end{bmatrix}$
5. Choose a 2×2 invertible key matrix K .
for example $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$
6. Encrypt using matrix multiplication modulo 26.
 $C = K \cdot P \text{ mod } 26$ for each plaintext vector P
7. Convert the resulting numbers back to letters to get the ciphertext.

A	B	C	D	E	F	G	H	I	J
1	2	3	4	5	6	7	8	9	10
K	L	M	N	O	P	Q	R	S	T
11	12	13	14	15	16	17	18	19	20
U	V	W	X	Y	Z				
21	22	23	24	25	26				

Modular Arithmetic Rules in Mod 26

Modular arithmetic (mod 26) is arithmetic where numbers "wrap around" after reaching 26 — perfect for the English alphabet (26 letters: A–Z). Here are the **basic rules**:

1. Addition Rule

$(a + b) \bmod 26 = \text{Remainder when } (a + b) \text{ is divided by } 26$
Ex. $(20 + 10) \bmod 26 = 30 \bmod 26 = 4$

2. Subtraction Rule

$(a - b) \bmod 26 = \text{Remainder when } (a - b) \text{ is divided by } 26$
Ex. $(5 - 9) \bmod 26 = -4 \bmod 26 = 22$

3. Multiplication Rule

$(a \times b) \bmod 26 = \text{Remainder when } (a \times b) \text{ is divided by } 26$
Ex. $(7 \times 5) \bmod 26 = 35 \bmod 26 = 9$

4. Inverse Rule

$$(a \times a^{-1}) \bmod 26 = 1$$

Ex. $5 \times 21 = 105 \bmod 26 = 1$

$\Rightarrow 21$ is the inverse of $5 \bmod 26$

Theorem and Example

Theorem 10.13.1

For any integer a and modulus m , let

$$R = \text{remainder of } \frac{|a|}{m}$$

Then the residue r of a modulo m is given by

$$r = \begin{cases} R & \text{if } a \geq 0 \\ m - R & \text{if } a < 0 \text{ and } R \neq 0 \\ 0 & \text{if } a < 0 \text{ and } R = 0 \end{cases}$$

Find the residue modulo 26 of **(a)** 87, **(b)** -38 , and **(c)** -26 .

Solution (a) Dividing $|87| = 87$ by 26 yields a remainder of $R = 9$, so $r = 9$. Thus,

$$87 = 9 \pmod{26}$$

Solution (b) Dividing $|-38| = 38$ by 26 yields a remainder of $R = 12$, so $r = 26 - 12 = 14$. Thus,

$$-38 = 14 \pmod{26}$$

Solution (c) Dividing $|-26| = 26$ by 26 yields a remainder of $R = 0$. Thus,

$$-26 = 0 \pmod{26}$$

Encryption: Example

Example 1. Use the key matrix $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$, to obtain the Hill cipher from the plain text “HELP”.

Solution: Step 1. Group successive plaintext letters into pairs and replace each plaintext letter by its numerical value. Here from the above table

$$H \rightarrow 8, E \rightarrow 5, L \rightarrow 12, P \rightarrow 16$$

Step 2 convert each plaintext pair into a column vector $P = \begin{bmatrix} H \\ E \end{bmatrix}$ plaintext vector.

Replacing each letter by its numerical value $P = \begin{bmatrix} 8 \\ 5 \end{bmatrix}$

Using formula for ciphertext vector

$$C = KP(mod 26) = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 8 \\ 5 \end{bmatrix} = \begin{bmatrix} 24 + 15 \\ 16 + 25 \end{bmatrix} = \begin{bmatrix} 39 \\ 41 \end{bmatrix} (mod 26) = \begin{bmatrix} 13 \\ 15 \end{bmatrix}$$

Now again from the table of Hill Ciphers

$$\begin{bmatrix} 13 \\ 15 \end{bmatrix} = \begin{bmatrix} M \\ O \end{bmatrix} \text{ ciphertext vector.}$$

Encryption: Example

Step 3

Now we work for “LP”

using numerical values of alphabets for the pair “LP”,

$$P = \begin{bmatrix} L \\ P \end{bmatrix} \text{ implies } P = \begin{bmatrix} 12 \\ 16 \end{bmatrix}$$

$$C = KP(\text{mod } 26) = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 12 \\ 16 \end{bmatrix} (\text{mod } 26) = \begin{bmatrix} 84 \\ 104 \end{bmatrix} (\text{mod } 26) = \begin{bmatrix} 6 \\ 0 \end{bmatrix}$$

Now

$$\begin{bmatrix} 6 \\ 0 \end{bmatrix} = \begin{bmatrix} F \\ Z \end{bmatrix} \text{ ciphertext vector.}$$

Hence, combining the ciphertext vectors the entire ciphertext message is **MOFZ**.

Deciphering/Decryption

The process of converting from cipher text to plain text is called deciphering.

$$\text{As, } C = K \times P$$

$$\text{So, } K^{-1} \times C = P$$

$$\begin{aligned} \text{And as,} \\ K^{-1} &= \frac{1}{|K|} \text{adj } K, \text{ mod } 26 \\ &= |K|^{-1} \times \text{adj } K \end{aligned}$$

Procedure to find $|K|^{-1}$

A number " a " has an inverse mod 26 if and only if:
 $\text{gcd}(a, 26) = 1$

This means:

- a and 26 must be **coprime** (they share no common factor except 1).
- If they are **not coprime**, then a has no modular inverse mod 26.

$\text{gcd}(4, 26) = 2 \rightarrow$ No inverse of 4

$\text{gcd}(13, 26) = 13 \rightarrow$ No inverse of 13

$\text{gcd}(3, 26) = 1 \rightarrow$ has inverse of 3

So... 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24 $\rightarrow$ No inverses

And... 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 $\rightarrow$ All are coprime to 26 $\rightarrow$ All have inverses

Deciphering/Decryption

Example 6. Decipher the Cipher text = MOFZ for Key matrix = $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$.

Solution: Recall the formula which we used for enciphering

$$C = K \times P \pmod{26}$$

$$P = K^{-1}C \pmod{26}$$

$$K^{-1} = \frac{Adj(K)}{\det(K)} \pmod{26}$$

$$\det(K) = |K| = \begin{vmatrix} 3 & 3 \\ 2 & 5 \end{vmatrix} = 15 - 6 = 9$$

$$\begin{aligned} K^{-1} &= \frac{\begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix}}{9} \\ &= 9^{-1} \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} \end{aligned}$$

What is the inverse of 9 modulo 26?

Let it be x , then $9x = 1 \pmod{26}$

$$9.1 = 9 \neq 1 \pmod{26}$$

$$9.2 = 18 \neq 1 \pmod{26}$$

$$9.3 = 27 = 1 \pmod{26}$$

Hence $9^{-1} = 3 \pmod{26}$

Deciphering/Decryption

$$K^{-1} = 9^{-1} \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} = 3 \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 15 & -9 \\ -6 & 9 \end{bmatrix} = \begin{bmatrix} 15 & 26-9 \\ 26-6 & 9 \end{bmatrix} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} (\text{mod } 26)$$

Step 2: Now, we will decipher MO first, then we will decipher FZ. For this we take

$$C = \begin{bmatrix} M \\ O \end{bmatrix} = \begin{bmatrix} 13 \\ 15 \end{bmatrix} \quad P = K^{-1}C (\text{mod } 26) = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 13 \\ 15 \end{bmatrix} = \begin{bmatrix} 450 \\ 395 \end{bmatrix} (\text{mod } 26) = \begin{bmatrix} 8 \\ 5 \end{bmatrix} \equiv \begin{bmatrix} H \\ E \end{bmatrix}$$

$$C = \begin{bmatrix} F \\ Z \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \quad P = K^{-1}C (\text{mod } 26) = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 90 \\ 120 \end{bmatrix} (\text{mod } 26) = \begin{bmatrix} 12 \\ 16 \end{bmatrix} \equiv \begin{bmatrix} L \\ P \end{bmatrix}$$

So all plain text is HELP

Encryption/Decryption

Numerical Value of Letters

<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>	<i>H</i>	<i>I</i>	<i>J</i>	<i>K</i>	<i>L</i>	<i>M</i>	<i>N</i>	<i>O</i>	<i>P</i>	<i>Q</i>	<i>R</i>	<i>S</i>	<i>T</i>	<i>U</i>	<i>V</i>	<i>W</i>	<i>X</i>	<i>Y</i>	<i>Z</i>
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	0

Mod 26 Multiplicative Inverse

<i>a</i>	1	3	5	7	9	11	15	17	19	21	23	25
<i>a</i> ⁻¹	1	9	21	15	3	19	7	23	11	5	17	25

Note:

For further exercise/problems

see word files uploaded to Portal

THANKS